

Kartik Raikar

AI Engineer — Generative AI — LLM Applications

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#) | [Atlas AI Resume](#)

Professional Summary

AI Engineer specializing in Generative AI, LLMs, RAG, and AI evaluation. Experienced in building production-oriented AI applications using Python, FastAPI, React.js, SQL, vector databases, and modern LLM tooling.

Projects

[Interactive AI Resume](#)

AtlasOS – AI Memory Operating System | Python, FastAPI, PostgreSQL, Redis, Qdrant, Next.js, Docker

- Built a **multi-tenant AI memory system** supporting **working, episodic, and semantic memory** across Redis, PostgreSQL, and Qdrant with asynchronous Celery pipelines.
- Enforced **cross-tenant isolation** via PostgreSQL RLS, tenant-scoped vector filtering, and immutable audit logs across relational and vector stores.
- Implemented **1024-dimensional semantic retrieval**, self-hosted BGE embeddings, and RoBERTa NLI contradiction detection with recency- and confidence-based conflict resolution.

ApexRAG – RAG Retrieval Evaluation System | Python, FastAPI, ChromaDB, Sentence Transformers, Ollama, scikit-learn

- Built a **RAG evaluation pipeline** over **2,580 React documentation chunks** and **100 human-verified Q&A pairs** using FastAPI on local CPU infrastructure.
- Benchmarked **5 retrieval strategies**, improving accuracy from **61% BM25** to **85% cross-encoder re-ranking**; identified ranking quality as the primary bottleneck.
- Resolved **data contamination, ID collisions, MDX parsing, and model drift** using corpus validation, path-based IDs, deterministic evaluation, and a Logistic Regression strategy router.

GitHub MCP Toolkit – Fault-Tolerant MCP Server | Python, FastMCP, GitHub API, Ollama, Docker, GitHub Actions

- Built an **MCP server** for LLM-driven GitHub issue management, semantic search, automated triage, repository operations, and reversible write workflows.
- Eliminated a measured **14% blind bulk-mutation rate** using a **two-phase SHA-256 preview-token protocol** requiring explicit human approval.
- Improved intent accuracy from **64% to 100%** and achieved **100% across 20 adversarial cases** using ABAC, circuit breakers, Saga rollback, prompt-injection isolation, and Pydantic validation.

RagaAI Catalyst – Enterprise LLM Observability Platform | Next.js, TypeScript, Tailwind CSS, Zustand, React Query, Recharts

- Built an **LLMOps observability dashboard** for tracing agentic workflows and analyzing **latency, token usage, execution traces, and model behavior**.
- Integrated **20+ LLM evaluation metrics** including faithfulness, hallucination, and context relevance with dataset-driven model analysis.
- Implemented **toxicity and prompt-injection guardrails**, automated red-teaming, dataset versioning, and synthetic-data generation using Next.js + TypeScript.

Education

B.E. Computer Science & Engineering (AI & ML)
Jain College of Engineering, Belagavi

Aug 2023 – Jul 2027
CGPA: 8.50/10

Pre-University Course (PCM)
Jain PU College, Belagavi

2021 – 2023
80%

Technical Skills

Programming: Python, JavaScript, TypeScript, SQL
Databases: PostgreSQL, MongoDB, Redis, Qdrant
AI/ML: Generative AI, LLMs, RAG, Transformers, AI Evaluation, Prompt Engineering, Semantic Search, Vector Search

Frameworks: FastAPI, React.js, Next.js, PyTorch
Tools: Git, GitHub, Docker, Linux, Postman, VS Code

Certifications

- Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate
- AWS Training & Certification – Fundamentals of Machine Learning and Artificial Intelligence
- Tata – GenAI Powered Data Analytics Job Simulation (Forage)

Leadership & Achievements

- Vice President – Department of Artificial Intelligence & Machine Learning, Jain College of Engineering, Belagavi:** Led departmental initiatives and coordinated technical and academic programs.
- Hack2Future 2.0 – IIIT Dharwad:** Competed as part of **Team Velora** in a national-level hackathon.
- Code for Innovex – NITTE NMAM Institute of Technology:** Participated in a **24-hour national-level hackathon**.

Relevant Coursework

Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks, Software Engineering, Artificial Intelligence, Machine Learning